Referential Regimes: Transformation-Based Identity Profiles for Neutral Substrates

Denise M. Case

Northwest Missouri State University, Computer Science and Information Systems, Maryville, MO, USA

Abstract

Data systems operating under persistent legal, institutional, and analytic disagreement require shared reference without presupposing agreement about interpretation. This paper investigates identity preservation under transformation, building on the conditional neutrality result of *Neutral Substrates*.

We pair candidate identity bases with families of record transformations and classify each family as preserving, breaking, neutral, or inapplicable for a basis. Using invariant-defined identity bases and explicit effect models, we derive and certify six classification cells in Lean 4 for scope decomposition, rule refinement, and forking. The derivations show how classifications depend on field granularity, admissible transformation steps, and explicit continuity warrants.

We also characterize the identity relations generated by invariant-preserving steps. Two bases generate equal relations exactly when they agree on the preservation of every admissible step in a shared comparison domain, and a single separating step witnesses a difference.

The formalization makes identity claims auditable, provides testable conditions for systems that transform records, and exposes the assumptions behind each classification. The results are conditional on the

declared bases, models, and warrants, whose adequacy for particular referents requires independent justification.

Keywords: formal ontology; identity profiles; transformation classification; individuation; co-reference; persistence; neutral substrates; referential common ground; stable reference

1 Introduction

Modern accountability systems increasingly operate under legal, institutional, and analytic disagreement. A shared record may need to support reference to the same intended subject while its representations are amended, forked, refined, decomposed, aggregated, or reclassified. Agreement about what a record means cannot be assumed, and neither can agreement about which identity conditions apply to its referents. This paper investigates candidate ways to characterize individuation, co-reference, and persistence under those transformations.

Neutral Substrates (Case 2026a) provides the motivating constraint: under Referential Common Ground and the stated foundational-layer restriction, a substrate satisfies semantic neutrality relative to its admissible frameworks. That conditional result does not establish a referential identity criterion for a particular bearer. The present paper examines what a declared criterion and selected transformation behavior can express, while keeping the adequacy of those choices open to independent justification.

The analysis pairs proposed identity bases with classifications of selected transformation families. An identity regime, in the terminology of this paper, is a declared basis together with its transformation profile. A classification records a proposed preservation, breakage, neutrality, or inapplicability judgment relative to a declared basis. Relations generated from admissible transformation steps are formal constructions; their interpretation as strict identity requires independent justification.

We select reference kinds for analysis: obligation-bearers, rules, occurrences, scopes, records, and plain referents. For each we propose candidate carrier–basis profiles. Some kinds receive paired bases that distinguish a locus from an object occupying it, a scope's coverage from its internal structure, and a rule's content from its structure. Forking, aggregation or decomposition, and refinement provide tests of those distinctions.

The candidate profiles have pairwise distinct classification functions, and the carrier–basis assignment is determinate. Whether distinct profiles generate distinct equivalence relations depends on the generating rule, the admissible transformation dynamics, and the comparison domain (Appendix A). A carrier-indexed combination holds separate relations within one explicitly tagged relation (Section 7).

The principal contribution is a formal method for deriving and examining identity-preservation classifications under transformation.

First, identity claims become auditable: a verdict follows from declared identity invariants, admissible transformation steps, and explicit assumptions that can be independently examined.

Second, the construction supplies testable conditions for record-transforming systems. An admissible transformation step that preserves one declared invariant but changes another provides a concrete witness of different generated identity relations.

Third, the formalization identifies which assumptions carry each result. The scope and rule examples expose field-granularity and transformation-domain dependencies; the fork example exposes the need for explicit continuity warrants.

The classifications, generated relations, and the distinctions among them are formal results about the declared bases and models. Applying them to a particular referent requires an independently justified identity criterion (Section 11).

The remainder of the paper proceeds as follows. Section 2 states the Neutral Substrate foundation and the narrower contribution pursued here. Section 3 sets out the formal setting and transformation classifications. Section 4 specifies the selected reference inventory. Section 5 develops conditional comparisons among identity bases, and Section 6 collects the candidate profiles. Section 7 compares their generated relations and shows how carrier-specific relations combine. Section 8 establishes determinacy of the selected profile assignment, while Section 9 treats application settings as identity-criterion tests. Section 10 situates the work in prior research; Section 11 states its boundaries; and Section 12 concludes.

2 Background: Neutral Substrates

This paper builds on *Neutral Substrates* (Case 2026a). That work studies a shared representational base intended to support reference under persistent interpretive disagreement without adopting contested causal or normative interpretations as unqualified foundational commitments. The present paper investigates proposed identity conditions and their behavior when representations are transformed. It does not assume that the abstract referential interfaces have been given adequate identity criteria for every intended kind of referent.

The definitions below retain their *Neutral Substrates* labels and wording. They establish the terminology imported for the present analysis.

2.1 Inherited Definitions

Definition 2.1 (Substrate)**.** A *substrate* $\mathcal{S}$ is a shared representational base providing stable reference for entities, occurrences, and institutional artifacts across a class of admissible interpretive frameworks $\mathbb{F}$. Stable reference consists of individuation, co-reference, and persistence.

A substrate is the foundational layer, not the complete record system. It can support later causal, normative, evidentiary, or institutional interpretations without adopting those interpretations as its own object-level commitments. The distinction between a referent and interpretations attributed to it is central; preserving a shared label or descriptive record does not itself establish that the intended referent has been fixed.

Definition 2.2 (Referential Regime)**.** A *referential regime* is the triple of

1. individuation conditions,
2. co-reference conditions, and
3. persistence conditions

by which a substrate fixes and tracks entities, occurrences, and institutional artifacts.

Individuation concerns what counts as one whole and sameness across presentations; co-reference concerns whether references denote the same referent; persistence concerns strict identity across settings and must be distinguished from succession, derivation, or counterpart relations. The abstract triple supplies interfaces for those questions, not their independently warranted answers. A relation defined by closure of proposed preserving steps does not, without additional grounding, establish identity in an intended domain.

Definition 2.3 (Referential Commitments)**.** Let $\mathcal{S}_{\text{ref}}$ denote the *referential commitments* of $\mathcal{S}$. These are the substrate-layer commitments fixed by its referential regime, including identifiers; the typing of entities, occurrences, and institutional artifacts; timestamps; provenance; and referential relations among them.

Identifiers, types, timestamps, provenance, and relations may supply materials for referential fixing. Their mere presence does not justify which

object-language propositions are fixed by a proposed referential regime. That connection requires independently grounded interpretation.

Neutral Substrates distinguishes a permitted attribution from the proposition attributed: accepting the attribution does not require accepting its content. It also defines interpretive non-commitment, extension stability, semantic neutrality, and neutrality by design.

Its central result is conditional sufficiency. Under Referential Common Ground, restricting foundational commitments to referential commitments and permitted attribution propositions suffices for semantic neutrality relative to every admissible framework. Permitted attributions are determined by the referential commitments.

The sufficiency result is conditional on Referential Common Ground and the stated restriction on foundational commitments. Its application requires independently justified referential criteria; the present paper inherits these obligations.

2.2 Contribution

This paper selects a finite reference inventory, candidate identity bases, and transformation families through which to investigate identity-sensitive distinctions. It studies classifications of re-expression, annotation, refinement, aggregation or decomposition, re-contextualization, reassignment, substitution, branching, provenance extension, and state evolution. The resulting classification profiles support formal comparison of candidate identity bases.

Definition 2.4 (Transformation-Invariant Identity)**.** Relative to a specified referent kind, proposed identity basis, and admissible transformation instances, a presentation is treated as *transformation-invariant* along steps classified as preserving the declared basis. A transformation classification expresses a proposed identity judgment relative to that basis; it does not, without independent justification, establish strict identity of the represented referent across those steps.

Section 3 formalizes candidate profiles and a relation generated from non-breaking steps; Section 5 compares selected basis-sensitive classifications; Section 7 compares generated relations and their combination; and Section 11 states the boundaries of the analysis.

3 Formal Setting

This paper investigates candidate conditions for stable reference under transformation in settings of persistent interpretive disagreement. Its formal results are relative to the selected carriers, identity bases, transformation families, and assumptions stated below.

3.1 Persistent Disagreement

We consider environments in which interpretive frameworks may remain in disagreement and their extensions need not be coordinated. Frameworks may advance incompatible causal accounts, normative attributions, or interpretive classifications without being required to converge. The neutrality constraint of Section 2 concerns which propositions the shared substrate itself adopts; it does not adjudicate among those interpretations.

This setting assumes neither that all admissible frameworks agree nor that they are mutually consistent. The conditional neutrality result of *Neutral Substrates* also requires Referential Common Ground: the complete adopted referential commitment set must remain jointly compatible with each admissible framework. That premise is not established by the present inventory or transformation classifications.

Revising a substrate merely to make conflicting interpretations converge would not resolve their disagreement neutrally. Revising an inadequately justified referential commitment is a different matter: any revised construction requires renewed examination of its own compatibility and referential

warrant. This paper does not define such a revision procedure.

3.2 Stable Reference in Strong Form

We examine the demanding case in which representations of a purported referent change through ordinary record transformations while the question of its identity remains meaningful. The referential regime in *Neutral Substrates* supplies abstract individuation, co-reference, and persistence interfaces; it does not independently determine which referents persist through a particular operation.

Stable reference requires identifying the intended referent and supplying an adequate criterion for when presentations concern that same referent. A change in location, role, description, or state need not constitute a change of bearer identity; conversely, preservation of a record or label does not establish the persistence of what it represents. The applicable judgment depends on the proposed identity criterion and its independent justification.

Roles, flags, contextual predicates, and carrier/basis parameters are not inherently incompatible with neutral shared reference. They may carry explicit, fixed distinctions. What matters is whether their meanings and use are adequately specified and whether the resulting referential commitments satisfy the relevant common-ground conditions. A convention whose identity-relevant meaning is left to conflicting interpretations cannot, without further grounds, establish a shared identity judgment.

3.3 Transformation Basis

Identity questions become visible when representations are transformed. We select the following finite transformation basis for analysis:

$$\mathcal{F} = \{\mathrm{RE}, \mathrm{AN}, \mathrm{RF}, \mathrm{AD}, \mathrm{RC}, \mathrm{RA}, \mathrm{SU}, \mathrm{BF}, \mathrm{PV}, \mathrm{SE}\}.$$

The basis is a selected set of operation families, not a claim of completeness for record systems or of identity effects uniform across all referent kinds.

Definition 3.1 (Transformation basis)**.** The transformation basis $\mathcal{F}$ consists of these operation families:

(a) RE, re-expression: change of representation format while preserving represented content;
(b) AN, annotation: addition of non-identity-bearing information;
(c) RF, refinement: increase in descriptive, structural, or normative detail;
(d) AD, aggregation or decomposition: restructuring into coarser or finer components under a specified whole-preservation condition;
(e) RC, re-contextualization: change in applicability context;
(f) RA, reassignment: change in association, bearer, or assignment relation;
(g) SU, substitution: replacement of the designated carrier;
(h) BF, branch or fork: divergence into multiple continuations;
(i) PV, provenance extension: addition of historical, derivational, or evidentiary metadata;
(j) SE, state evolution: change in state of an enduring referent over time.

These names designate typed operation families. Their meaning and identity effects must be evaluated relative to an intended referent, eligible presentations, and a stated identity basis. In particular, AD may preserve the covered cases of a scope while changing its internal organization; whether either property is identity-relevant is a separate question. Changes to covered cases are not represented by that particular whole-preserving instance of AD.

If a larger transformation basis preserves the previously assigned classifications on $\mathcal{F}$, then any existing classification disagreement remains a disagreement on the larger basis, as expressed in Proposition 7.2.

3.4 Classification of Transformations

The construction assigns a classification to each operation family relative to a declared identity basis and its intended comparison domain.

Definition 3.2 (Transformation classification)**.** For a profile τ and transformation family $f \in \mathcal{F}$,

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{BRK}, \mathsf{NEU}, \mathsf{N/A}\}.$$

The proposed classifications mean:

(a) PRS: the operation preserves the declared identity basis;
(b) BRK: the operation breaks the declared identity basis;
(c) NEU: the operation is applicable but does not affect that basis;
(d) N/A: the operation is inapplicable on the selected domain.

The four-valued classification above describes the stipulated profile catalog. The canonical Lean matrix uses three classification values (IGN, PRS, BRK), mapping both NEU and N/A to IGN. The companion Persistence classification restores applicability at the N/A cells of Table 1.

The newer invariant-based construction instead derives one of five verdicts: N/A, BRK, MIX, PRS, or NEU. It also distinguishes a modeled N/A verdict from a matrix cell for which no derivation has been supplied.

These representations are not interchangeable without an explicit semantic correspondence.

PRS and NEU both designate non-breaking behavior in this classification scheme, but they express different reasons. A family-level value is meaningful only when its proposed behavior is sufficiently uniform over the specified eligible operations; otherwise the intended scope must be refined. Neither the classification label nor its recorded presence in a profile independently warrants strict identity for particular referents.

Definition 3.3 (Non-breaking transformation)**.** A family f is *non-breaking*

for profile τ when

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{NEU}\}.$$

It is *breaking* when $\mathrm{Class}_\tau(f) = \mathsf{BRK}$.

3.5 Identity Regimes and Generated Relations

An identity regime is a proposed pairing of an identity basis with a transformation classification. It is not itself an ontological kind or proof that its basis is adequate for a particular referent.

Definition 3.4 (Identity regime)**.** An *identity regime* τ consists of a declared identity basis and a classification function

$$\mathrm{Class}_\tau : \mathcal{F} \to \{\mathsf{PRS}, \mathsf{BRK}, \mathsf{NEU}, \mathsf{N/A}\}.$$

The basis proposes conditions of sameness for the intended referent; the classification records proposed transformation behavior relative to that basis.

Definition 3.5 (Induced relation from non-breaking families)**.** Fix a profile τ and a domain D_τ of eligible presentations. Let $E_\tau \subseteq D_\tau \times D_\tau$ consist of pairs joined by specified, admissible steps of transformation families f satisfying

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{NEU}\}.$$

Define $\sim_\tau$ as the smallest equivalence relation on D_τ containing E_τ.

This is a classification-based family-level construction. It selects admissible steps according to the stipulated family classifications; it does not independently establish that each step preserves the declared identity basis.

The companion Persistence construction defines a different family-level relation, generated only by steps of families classified PRS. Under common domains and transformation dynamics, the PRS-only generated relation is contained in the relation generated by PRS and NEU families.

The newer invariant-based construction instead evaluates preservation at the level of individual admissible steps. Its generating edges are exactly the steps that preserve the declared invariant, including preserving steps of families whose derived verdict is MIX. The three constructions must not be conflated.

Each generated relation is an equivalence relation by construction. None independently establishes strict identity of the intended real-world referents. Such an interpretation requires an independently justified identity criterion, evidence supporting the admissible steps, and justification that equivalence closure preserves the criterion.

In particular, branching or a chain of classified non-breaking steps cannot, by itself, discharge these scientific obligations.

3.6 Basis Plurality

A carrier kind may admit more than one proposed identity basis. For example, one can compare locus-fixed with object-fixed criteria, coverage with structural organization, or rule content with rule structure. These comparisons are well defined whichever basis an application adopts.

Definition 3.6 (Basis plurality)**.** A carrier kind exhibits *basis plurality* within a construction when at least two distinct identity bases are specified for it. Whether both bases suit a given application is a separate question.

The selected construction examines pairs of bases for plain referents (locus or object), scopes (extension or structure), and rules (content or structure).

3.7 Hidden Regimes

A representation can fail to disclose an identity-relevant distinction on which its identity judgments nevertheless depend. This issue is separate from whether those judgments are expressed in one relation or several relations.

Definition 3.7 (Hidden regime)**.** A *hidden regime* is an identity-relevant criterion or distinction used to determine sameness in a realization but not adequately declared or fixed in the shared referential account, so that its application depends on an unstated or contested interpretation.

A declared role, flag, carrier tag, or basis parameter is not hidden merely because it is a discriminator. Its adequacy depends on whether its semantics are fixed and its use is justified for the intended referents. Conversely, an unacknowledged convention cannot be treated as established common ground merely because a system implements it. Whether a proposed combined representation is faithful must therefore be examined rather than decided by terminology alone.

3.8 Scope of the Analysis

The selected carrier inventory, identity bases, and transformation classifications supply a finite candidate construction. Sections 5–7 analyze its transformation distinctions and the relations they generate. The construction does not supply a complete inventory of identity criteria or a general procedure for determining strict identity under transformation.

The next section presents the selected reference inventory. Its carrier kinds identify subjects of inquiry; they do not themselves supply independent identity warrants.

4 Reference Inventory

Before analyzing identity under transformation, we select a finite inventory of referents encountered in the accountability settings considered in this paper. These settings call for descriptions of parties, rules, occurrences, scopes, records, and things acted upon or observed. The inventory organizes questions of reference without adopting a contested causal or normative

interpretation as a substrate commitment.

The inventory is a working selection, not a theorem, an upper ontology, or a complete domain ontology. The kinds need not be mutually exclusive, and their names do not by themselves supply criteria of individuation or persistence.

Reference kinds organize the intended referents; identity bases propose conditions for their sameness under transformation.

4.1 Reference Kinds and Identity Carriers

Definition 4.1 (Reference kind)**.** A *reference kind* is a class of intended referents selected to organize questions of reference in the accountability settings under analysis. Membership in a reference kind does not itself determine a referent's unity, sameness, or persistence criteria.

Definition 4.2 (Identity carrier)**.** An *identity carrier* is an intended referent considered as the subject of a proposed sameness, difference, or persistence judgment under transformation. The designation does not itself warrant that judgment or determine the applicable identity basis.

A reference kind can organize several proposed identity bases. Different kinds may also support compatible criteria or share referents considered in different capacities. In either case, the adequacy of an identity judgment depends on an identified bearer and an independently justified criterion, not solely on a kind name or transformation label.

The selected inventory is:

Reference kind	Referent considered in this analysis
Obligation-bearer	a party considered as a possible bearer of responsibility, duty, liability, or authority
Rule	a rule or normative instrument considered separately from an act of enacting, interpreting, or applying it
Occurrence	a happening such as a filing, inspection, transfer, or decision, whose unity and identity require specified criteria
Scope	a domain of applicability described by places, parties, periods, procedures, classifications, or conditions
Record	a document, entry, or other informational record that can be referred to independently of what it describes or asserts
Plain referent	a site, object, facility, instrument, sample, or asset considered apart from any assigned obligation-bearing or rule-governing role

These categories draw on distinctions familiar in foundational ontology, legal ontology, provenance, and institutional recordkeeping. No novelty or independent necessity is claimed for this inventory. Its role is to provide candidate carriers for the subsequent analysis.

4.2 Obligation-Bearers

The selected settings include claims about parties that may bear responsibility, duty, liability, authority, or accountability. The substrate can preserve reference to a party and record that a source attributes an obligation to it without adopting that attribution as an unqualified normative conclusion.

In multi-jurisdiction regulatory reporting, frameworks may identify different

entities or capacities as relevant: a parent organization, subsidiary, facility, office, or delegated agent. The organizational entity, an institutional role, and the occupant of that role need not have identical persistence criteria. Whether two presentations concern the same bearer requires an account appropriate to the intended referent.

Distinguishing a bearer from occurrences in which obligations are assigned, contested, fulfilled, or discharged supports comparisons across time. One bearer may participate in several occurrences, and one occurrence may involve several bearers. These observations motivate separate reference questions.

4.3 Rules

The selected settings include reference to rules and normative instruments that may be enacted, amended, delegated, interpreted, applied, challenged, or repealed. Recording such a referent does not require the substrate to declare that the rule is valid, binding, justified, or correctly interpreted.

Legislative history analysis distinguishes questions about a statute from questions about the enactment, amendment, agency action, or judicial decision associated with it. A statute may remain citable across those occurrences, but citation continuity alone does not settle whether successive texts, versions, or normative contents constitute the same rule.

The proposed content-fixed and structure-fixed bases examine different responses to refinement. Whether either basis supplies a warranted identity criterion depends on which rule referent is intended and what the transformation actually preserves.

4.4 Occurrences

The selected settings include occurrences such as filings, inspections, votes, transfers, transactions, observations, decisions, enactments, amendments, and enforcement actions. Their temporal realization and provenance can help

identify them but do not necessarily determine where one occurrence begins or ends, which phases belong to one whole, or when two reports concern the same happening.

For example, jurisdictions may disagree about whether a filing constitutes compliance while referring to an occurrence they have independently identified as the same filing. Similarly, institutions may dispute the significance of an enactment or a measurement without thereby disputing its occurrence identity. In other cases, that identity itself may be contested and cannot simply be assumed as common ground.

An occurrence can be distinguished for analysis from its participants, the records describing it, and the rules invoked in its evaluation. Such analytic distinctions do not by themselves establish disjoint ontological domains.

4.5 Scopes

The selected settings include applicability domains described by places, parties, periods, procedures, classifications, or conditions. Scopes may be territorial, institutional, temporal, procedural, classificatory, jurisdictional, or conditional. Their boundaries and relevance may themselves be disputed.

In multi-jurisdiction reporting, a rule may be associated with different entities, facilities, transactions, or reporting periods. Tracking a scope separately in an analysis can permit comparison of coverage while holding the rule description fixed, or comparison of rule versions while holding coverage fixed. This is a useful modeling choice, not proof that every substrate must make scope an independent identity carrier.

The proposed extension-fixed and structure-fixed bases distinguish coverage from internal organization. A decomposition may preserve covered cases while changing organization; whether it preserves identity depends on the justified scope criterion, not solely on the operation's name.

4.6 Records

The selected settings include descriptive documents, dataset entries, observation records, provenance items, and other informational records. A substrate may record that a source asserted a claim without thereby adopting its causal or normative content.

In cross-institutional provenance, a record may be produced, received, annotated, copied, transferred, or reanalyzed. The institutions may disagree about what it shows. Whether successive presentations are of the same record nevertheless depends on the identity basis: content, a particular token, a version, or a custody-linked object may give different answers.

Distinguishing a record from what it describes permits separate questions about an assertion and its subject. It does not imply that annotation, copying, or transfer always preserves record identity.

4.7 Plain Referents

The selected settings also include things acted upon, regulated, located, transferred, observed, measured, or described: facilities, parcels, instruments, devices, environmental sites, physical objects, infrastructure components, samples, and assets.

The designation *plain referent* indicates how a referent is being considered in the present analysis, not a claim that the referent could never bear an obligation, occupy a role, or appear in another reference kind. A facility may be inspected; a parcel may fall within an applicability scope; an instrument may produce a measurement record; and a sample may pass between institutions. Whether the referent persists through these operations remains a separate question.

The proposed locus-fixed and object-fixed bases distinguish questions about a place from questions about an object associated with that place. A branch

or fork may provide a useful test of their different transformation responses, but does not by itself establish that they are two identity criteria for one and the same kind of bearer. The intended referents and eligible operations must be specified.

4.8 Summary

The selected reference kinds organize a set of candidate identity questions. The distinction between a selected carrier and its proposed identity basis remains essential: the inventory identifies intended referents, and the bases propose criteria whose consequences can be tested under transformation.

The next section compares proposed identity bases and transformation behavior for these selected carriers under the formal setting of Section 3.

5 Transformation-Invariant Identity

The selected reference inventory of Section 4 identifies candidate carriers for comparing identity judgments under transformation. This section examines how declared identity bases and classification profiles can differ over the transformation basis of Section 3. A classification records proposed behavior relative to a specified basis; it does not independently establish that two presentations have the same real-world referent. The scientific adequacy of the basis, the operation, and their interpretation remains a separate obligation.

The inquiry draws on the distinction between reference kinds and identity conditions familiar in formal ontology (Guarino 1999; Guarino and Welty 2002). A kind name alone does not settle whether a transformation preserves its intended bearer. A stated identity basis helps formulate a test, but a warranted persistence judgment also requires evidence and an appropriate criterion for that referent.

5.1 Identity Bases

Definition 5.1 (Identity basis)**.** An *identity basis* is a stated criterion proposed for deciding when eligible presentations concern the same intended referent. It specifies the conditions whose preservation would warrant sameness under the interpretation in question. Its declaration alone does not establish that the criterion is adequate for the intended referent.

The selected construction considers single candidate bases for obligation-bearers, occurrences, and records, and paired bases: content versus structure for rules, extension versus structure for scopes, and locus versus object for plain referents. These are proposed criteria for examining transformation behavior.

A transformation classification is meaningful only relative to the identity basis, the eligible comparison domain, and the actual effects of the operation. Different candidate bases may classify a selected operation differently; that does not by itself decide which basis is appropriate or whether the resulting judgments warrant strict identity.

5.2 Profiles and Collapse

Section 3 defines identity regimes as proposed basis–classification profiles and defines an equivalence relation $\sim_\tau$ generated from specified non-breaking steps. The generated relation and an independently warranted strict-identity relation must not be conflated.

Definition 5.2 (Collapse)**.** Fix a common comparison domain D of eligible presentations for two profiles τ and τ'. The profiles *collapse at the level of generated relations on* D when the restrictions of their generated relations agree:

$$\sim_\tau\big|_D = \sim_{\tau'}\big|_D.$$

Those generated relations are *distinct on* D if one relates an eligible pair

that the other does not.

Distinct classification functions do not necessarily induce distinct generated relations. For example, a family assigned PRS under one profile and NEU under another is included under both profiles in the classification-based PRS-or-NEU construction. On common domains and transformation dynamics, its admissible steps therefore contribute generating edges to both relations.

Those stipulated assignments do not independently establish that every actual step preserves both declared invariants.

In the older family-generated construction, which uses only families classified PRS, distinct sets of preserving families need not yield distinct relations on arbitrary transformation dynamics, because the distinguishing families may have no relevant steps. In the Persistence construction, profiles with identical preserving sets generate identical relations on every shared dynamics (Appendix A).

A stronger result holds in the invariant-based step construction: when an admissible step inside a common comparison domain preserves one basis invariant but changes another, the two step-generated relations differ. The endpoints of the separating step cannot be connected by any path of steps that each preserve the second basis, because equality of invariant values is preserved along such a path (Proposition 6.5). This is a mathematical fact about the specified invariants and admissible steps, not independent scientific warrant for strict identity of the intended referents.

Definition 5.3 (Underdetermined kind)**.** A reference kind is *underdetermined at a transformation family* f within the selected construction when two proposed bases for that kind assign, on a shared eligible comparison domain, a non-breaking classification and a breaking classification, respectively, to f.

This is a difference between candidate classification profiles. If both bases are adequately justified for a particular use, a system must retain the distinction between the judgments they authorize. It does not follow that they require

separately implemented or non-amalgamable relations: an explicit basis parameter may preserve both. Nor does a classification difference alone establish a separating pair in the generated equivalence relations or in the world.

5.3 Carrier Kinds Not Split in the Core Construction

The selected catalog specifies one candidate basis for each of obligation-bearer, occurrence, and record.

For an *obligation-bearer*, a possible criterion tracks the continuing party rather than each obligation attributed to that party. A change in location, assigned role, or obligation may preserve that party, provided an appropriate account of the party's continuity warrants the judgment. Replacement of the party may instead break its identity. Organizational continuity and continuity of role occupancy must not be conflated.

For an *occurrence*, temporal realization and provenance may supply useful evidence, but they do not by themselves establish one bounded whole. Redescription of an independently individuated occurrence may preserve it; treating a proper phase as the whole may not. Decomposition, branching, and other transformations cannot be classified uniformly without specifying the occurrence and eligible presentations.

For a *record*, content identity, token identity, version identity, and continuity of custody can produce different judgments. Copying a report may preserve its content while creating another token; annotation may preserve an informational core while creating a new version. Neither preservation nor breakage follows from the label "record" alone.

The selected profile names are

OBL, OCC, REC.

Their intended identity criteria and transformation classifications remain

subject to the qualifications above.

5.4 Witnessed Split Kinds

The other carrier kinds in the selected catalog have pairs of candidate bases. Their comparisons can be expressed by a common classification pattern. In particular, a difference between proposed values of $\mathrm{Class}_\tau(f)$ is an algebraic fact about the profiles; a difference between warranted identity relations requires more.

Lemma 5.4 (Split pattern for classification profiles)**.** *Let K be a reference kind and let τ_1, τ_2 be two candidate profiles for K over the same transformation basis and a shared eligible comparison domain. If some $f \in \mathcal{F}$ satisfies*

$$\mathrm{Class}_{\tau_1}(f) \in \{\mathsf{PRS}, \mathsf{NEU}\} \quad \textit{and} \quad \mathrm{Class}_{\tau_2}(f) = \mathsf{BRK},$$

then the classification functions of τ_1 and τ_2 are distinct. This conclusion concerns the classification functions; the relations they generate are compared in Section 7.

Proof. The first classification is different from BRK, so the classification functions disagree at f and cannot be equal. Their generated relations depend on the eligible step pairs and equivalence closure; those relations, and their interpretations, are not determined by this disagreement alone. □

A stronger, referent-level separation would need a common eligible pair of presentations together with independently justified identity criteria that agree and disagree on that pair. The following examples show how candidate separating situations can be constructed. They do not prove that the corresponding criteria are necessary for every realization of the reference kind.

5.5 Scope Under Decomposition

For a scope presentation, distinguish the cases covered from their internal organization. Consider a finite set $C = \{a, b\}$ and two partitions

$$P = \{\{a, b\}\}, \qquad P' = \{\{a\}, \{b\}\}.$$

The presentations (C, P) and (C, P') have the same extension but different partition structures. A stipulated extension-fixed basis identifies them; a stipulated structure-fixed basis distinguishes them. An eligible decomposition that replaces P by P' while preserving C therefore provides a candidate separating example.

Proposition 5.5 (Scope underdetermination in the selected model)**.** *If an eligible* AD *operation preserves the coverage C while changing the identity-relevant partition P, then the extension-fixed and structure-fixed candidate profiles can assign different classifications to that operation.*

Proof. Under the stipulated extension-fixed criterion, equality of C supplies preservation; under the stipulated structure-fixed criterion, inequality of P supplies breakage. On a domain where these are the specified effects of AD, its two classifications differ, as in Lemma 5.4. □

We call the two proposed profiles

$$\text{SCOPE-E}, \qquad \text{SCOPE-S}.$$

This example demonstrates a distinction between coverage and structure within the model. It does not show that partition equality is the correct identity criterion for every scope, or that a scope cannot be faithfully represented with an explicit basis parameter.

5.6 Rule Under Refinement

For a rule presentation, distinguish the specified normative content from its textual or organizational structure. Consider two presentations (N, S) and (N, S') with identical stipulated content N but different structures $S \neq S'$. A content-fixed comparison identifies the presentations, whereas a structure-fixed comparison separates them. Whether a particular refinement actually preserves normative content must be assessed rather than inferred from the word "refinement."

Proposition 5.6 (Rule underdetermination in the selected model)**.** *If an eligible* RF *operation preserves the specified content N while changing the identity-relevant structure S, then the content-fixed and structure-fixed candidate profiles can assign different classifications to that operation.*

Proof. The stipulated content-fixed criterion is preserved because N is unchanged; the stipulated structure-fixed criterion is broken because $S \neq S'$. The corresponding classifications differ, as in Lemma 5.4. □

We call the two proposed profiles

RULE-C, RULE-S.

This comparison does not decide whether a particular statute or rule persists across amendment, recodification, or re-expression. That requires a justified account of the relevant rule referent.

5.7 Plain Referent Under Forking

For plain referents, distinguish a locus from an object associated with that locus. A toy configuration (L, O) records both. Two configurations (L, O_1) and (L, O_2) may have the same locus L but different objects $O_1 \neq O_2$. A locus-fixed comparison then identifies them, while an object-fixed comparison

distinguishes them. The inequality of the objects is an assumption of the toy model, not a conclusion from their labels alone.

Proposition 5.7 (Plain-referent underdetermination in the selected model)**.** *If an eligible* BF *operation yields compared presentations with the same locus but independently distinct objects, then locus-fixed and object-fixed candidate profiles can assign different classifications to that comparison.*

Proof. The locus-fixed criterion is preserved by the shared locus; the object-fixed criterion is broken by the stipulated object distinction. The candidate classifications consequently differ, as in Lemma 5.4. □

We call the two proposed profiles

LOC, OBJ.

A fork of record representations does not by itself create distinct objects or imply loss of object identity. Nor does relocation of an object necessarily end its identity. The separating case depends on which referent is being tracked and whether the assumed operation and object distinction are independently warranted.

5.8 Result of the Algebra

The selected inventory contributes single-basis candidate profiles for obligation-bearers, occurrences, and records, and paired profiles for plain referents, scopes, and rules:

OBL, OCC, REC,
LOC, OBJ, SCOPE-E, SCOPE-S, RULE-C, RULE-S.

The conditional examples show how different identity bases can classify specified transformations differently. They illustrate tests of the bases; they

do not establish that every transformation of a named family has the same effect.

Section 6 collects the proposed classifications; Section 7 compares the relations they generate and shows how carrier-specific relations combine.

6 Candidate Carrier–Basis Profiles

Sections 4 and 5 yield a catalog of named carrier–basis profiles, selected for analysis.

The following labels name the selected profiles:

OBL	obligation-bearer identity.
OCC	occurrence identity.
REC	record identity.
LOC	locus-fixed plain-referent identity.
OBJ	object-fixed plain-referent identity.
SCOPE-E	extension-fixed scope identity.
SCOPE-S	structure-fixed scope identity.
RULE-C	content-fixed rule identity.
RULE-S	structure-fixed rule identity.

Table 1 records proposed classifications over the finite transformation basis $\mathcal{F}$. Its entries are stipulated family-level profile assignments for the selected comparison settings. They must not be read as established identity effects of every operation in the named family or as independently warranted judgments about real-world referents. Where eligible operations within a family have different effects, that family requires a more specific classification domain.

	OBL	OCC	REC	LOC	OBJ	SCOPE-E	SCOPE-S	RULE-C	RULE-S
RE	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU
AN	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
RF	PRS	PRS	PRS	PRS	PRS	PRS	BRK	PRS	BRK
AD	PRS	BRK	PRS	PRS	PRS	PRS	BRK	PRS	BRK
RC	NEU	NEU	NEU	NEU	NEU	PRS	BRK	NEU	NEU
RA	NEU	N/A	NEU	N/A	N/A	NEU	BRK	NEU	NEU
SU	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK
BF	NEU	N/A	PRS	PRS	BRK	PRS	BRK	BRK	BRK
PV	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
SE	NEU	PRS	NEU	PRS	PRS	N/A	N/A	N/A	N/A

Table 1: Proposed classification profiles over $\mathcal{F}$ for the selected carrier–basis construction. PRS: preservation of the declared basis; BRK: breakage of that basis; NEU: applicable but identity-neutral under that basis; N/A: inapplicable on the selected domain. Entries are candidate assignments, not universal identity laws.

A PRS entry proposes preservation of the declared identity basis; a BRK entry proposes its breakage. An NEU entry indicates an applicable operation that does not affect the basis, and an N/A entry indicates inapplicability on the selected domain. Neither PRS nor NEU independently warrants strict identity between actual referents. In particular, a BRK entry in a proposed profile does not by itself establish that two specified objects are distinct.

For the selected same-carrier comparisons, BF distinguishes the proposed LOC and OBJ classifications; AD distinguishes SCOPE-E and SCOPE-S; and RF distinguishes RULE-C and RULE-S. The conditional examples in Section 5 show how stipulated locus, object, extension, structure, and content criteria can yield these classification differences. The other entries record additional proposed behavior whose adequacy also requires examination; they are not needed to establish the conditional split comparisons.

The uniform RE and SU rows are features of this selected table: RE is assigned NEU throughout, while SU is assigned BRK throughout. Those assignments are not general theorems about re-expression or substitution under every scientifically adequate identity criterion.

A disagreement between two classification functions is distinct from a dis-

agreement between the equivalence relations generated by non-breaking steps. To state the latter, the two relations must be compared on a common eligible domain.

Definition 6.1 (Separation witness)**.** Let τ and τ' be profiles with a common domain D of eligible presentations. A *separation witness for their generated relations on* D is a pair $(r, s) \in D \times D$ for which

$$r \sim_\tau s \quad \text{and} \quad r \not\sim_{\tau'} s,$$

or conversely, with each relation restricted to D.

Such a witness demonstrates that the generated relations differ on D; it does not independently establish the correctness of either relation as strict real-world identity. Nor does a difference in the classification of a transformation family necessarily provide such a witness, since equivalence closure may connect the relevant pair by another non-breaking path.

Proposition 6.2 (Distinct core profiles)**.** *The nine classification functions specified by the columns of Table 1 are pairwise distinct.*

Proof. It is enough to inspect the entries at AN, RF, RA, and BF. The four-entry classification signatures in these rows are pairwise different across all columns of Table 1. Hence no two of the classification functions agree on all of $\mathcal{F}$. □

This proposition concerns the *declared classification functions*. Whether their generated relations are pairwise distinct on a common domain is taken up in Section 7 and Appendix A.

6.1 Initial Model-Derived Classification Results

Table 1 records the proposed classification matrix. Its entries are not inputs to the following derivations.

Instead, the companion Lean models (Case 2026b) specify presentation fields, identity invariants, admissible transformation steps, and their effects.

An invariant basis consists of a function on presentations and a declared support of presentation fields. Agreement on the supported fields implies equality of the invariant. The Lean development additionally proves minimality of the supports used in the certified cells: every supported field can independently affect its invariant.

An effect model declares admissible transformation steps and the fields each transformation family may write. An effective instance is an admissible step that changes at least one presentation field.

Definition 6.3 (Model-derived transformation verdict)**.** Fix an invariant basis b, an effect model M, and a transformation family f.

The derived verdict is:

(a) N/A: no effective instance exists;
(b) BRK: effective instances exist and every one changes the invariant;
(c) MIX: some effective instance changes the invariant and another preserves it;
(d) PRS: effective instances exist, all preserve the invariant, and at least one changes a supported field;
(e) NEU: effective instances exist, all preserve the invariant, and none changes a supported field.

MIX establishes that the declared transformation family contains both invariant-preserving and invariant-breaking effective instances.

The family therefore cannot receive a uniform preserving or breaking verdict under those modeled conditions. A uniform verdict would require restricting or refining the family.

Alternatively, individual preservation judgments can be retained at the step level without assigning one uniform verdict to the family.

Applicability is defined for the model's presentation space, not for an arbitrarily selected smaller comparison domain.

Pair and family	Model or explicit warrant	First	Second
SCOPE-E/SCOPE-S, AD	Coverage-preserving; one organization field	PRS	BRK
SCOPE-E/SCOPE-S, AD	Coverage-preserving; field per case-pair	NEU	BRK
SCOPE-E/SCOPE-S, AD	Broader variant allowing coverage changes; field per pair	MIX	BRK
RULE-C/RULE-S, RF	Norm-preserving strict refinement; coarse organization field	PRS	BRK
RULE-C/RULE-S, RF	Norm-preserving strict refinement; fields per case-pair and case norm	NEU	BRK
RULE-C/RULE-S, RF	Broader variant permitting changes to norms	MIX	BRK
LOC/OBJ, BF	Branch changes; no further warrant	MIX	MIX
LOC/OBJ, BF	Colocated and FreshObject warrants	NEU	BRK
LOC/OBJ, BF	Colocated and SameObject warrants	NEU	NEU

Table 2: Selected verdicts under explicitly specified Lean effect models and warrants. The two rightmost columns follow the order of the bases named in the first column. The field-level strict AD, strict RF, and warranted BF rows account for the certified cells in the derived matrix. The remaining rows are supplementary model comparisons. All verdicts are conditional mathematical results, not universal identity laws.

The certified cells in the derived matrix of the companion Lean development are:

- SCOPE-E under strict AD: NEU;
- SCOPE-S under strict AD: BRK;
- RULE-C under strict RF: NEU;
- RULE-S under strict RF: BRK;
- LOC under BF with Colocated and FreshObject warrants: NEU;
- OBJ under BF with those same warrants: BRK.

Each certified cell contains an explicit effect model, an invariant basis with a

proven minimal support, and a proof of its derived verdict. The remaining cells in the stipulated matrix have no derived model yet. For every certified cell, the derived and stipulated classifications agree on whether the transformation is breaking.

The derived NEU cells are classified PRS in Table 1. Both verdicts are non-breaking, but they distinguish whether a transformation engages a supported field.

The scope and rule comparisons expose dependencies on field granularity. The fork classification additionally depends on explicit continuity warrants. These conditional results do not establish a universal correspondence between stipulated and derived classifications.

The additional rows in Table 2 are sensitivity analyses or alternative warranted models, not additional certified matrix cells.

Two findings are especially consequential. First, refining the scope presentation from one organization field to one field per case-pair changes the strict AD extension verdict from PRS to NEU. The fine-grained extension basis reads diagonal entries (coverage), while restructuring changes only off-diagonal entries. The support of each fine-grained scope basis is proved minimal in that model. Both verdicts are non-breaking, but the classification is not invariant under the choice of field representation. The model therefore exposes a granularity dependence that must be controlled before an unqualified classification can be asserted.

The rule model exhibits a corresponding dependence. Under strict RF, the content basis is PRS when the organization is represented coarsely, but NEU when fields separately represent case-pair organization and case-specific norms. The structure basis is BRK in both representations. The field-level content support reads diagonal organization entries and the case-specific norms; strict RF preserves both. The content and structure supports are proved minimal in this model.

Second, branching alone specifies a change of branch, not a change of locus

or object. With no additional warrant, both corresponding verdicts are MIX. Under the Colocated and FreshObject warrants, the locus basis is neutral and the object basis breaking; under the Colocated and SameObject warrants, both are neutral and their step-generated relations collapse on every comparison domain. These outcomes are consequences of distinct, explicitly stated hypotheses. None establishes that one warrant is true of every fork in an actual system.

The broader scope variant in the table permits changes to coverage and therefore is not an instance of the coverage-preserving AD family as originally defined in Section 3. It is included as a separate sensitivity analysis. The broader rule variant likewise changes the specification of eligible refinements. A comparison with Table 1 is warranted only after matching operation semantics, identity basis, field granularity, and admissibility conditions.

The invariant-based step construction uses a presentation space C, an effect model M, an invariant basis b, and a comparison domain $D \subseteq C$.

Define the preserving-step relation by

$$R_{M,b,D}(x,y) \iff x,y \in D \ \wedge \ \exists f \, \big(M.\mathrm{step}(f,x,y) \wedge b.\mathrm{inv}(x) = b.\mathrm{inv}(y)\big).$$

The step-generated identity relation is

$$\mathrm{StepId}_{M,b,D} = \mathrm{EqCl}_C(R_{M,b,D}),$$

where EqCl_C is the smallest equivalence relation on C containing the specified generating steps.

Unlike the classification-based family-level constructions, this relation tests invariant preservation for each admissible step. Its generating edges may include preserving steps from families whose derived verdict is MIX.

Theorem 6.4 (Step-generated relation equality)**.** *Fix a common effect model, two invariant bases b and b', and a comparison domain D.*

The equivalence relations generated by their respective invariant-preserving admissible steps are equal if and only if the bases agree on invariant preservation for every admissible step with both endpoints in D.

Proof. If preservation judgments agree on every admissible step in D, the generating relations are equal. Their equivalence closures therefore agree.

Conversely, suppose the generated relations agree. An admissible step preserving b belongs to the relation generated by b and hence also to the relation generated by b'.

Every generating step for the latter relation preserves b'. Equality of invariant values is an equivalence relation, so its closure also preserves b'. Thus the original step preserves b'. The converse follows symmetrically. □

Proposition 6.5 (Separating transformation step)**.** *Let $x, y \in D$ be connected by an admissible transformation step.*

If $b(x) = b(y)$ but $b'(x) \neq b'(y)$, then the step-generated identity relations for b and b' are distinct.

Proof. The step generates a relation between x and y under b.

Under b', the generated relation cannot connect x and y, because every generating step preserves b', and equality of invariant values is preserved under equivalence closure. □

These derivations are mathematically conditional.

Independently justifying the invariants and fork warrants for actual referents, and showing that a generated equivalence relation expresses warranted strict identity, remain separate scientific obligations.

7 Profile Distinctions and Carrier-Indexed Combination

The preceding sections specify candidate carrier kinds, identity bases, and transformation classifications. This section compares the resulting profiles by their classification functions and by the relations they generate, and shows how carrier-specific relations can be combined in one tagged relation.

7.1 Witnessed Differences Within the Selected Profiles

The proposed core contains pairs of identity bases associated with the same carrier kind. These pairs motivate tests of whether ordinary transformations make different identity judgments under the respective bases.

For plain referents, the proposed locus-fixed and object-fixed bases are compared using branch/fork operations (Proposition 5.7). A particular fork may preserve a locus while distinguishing the objects associated with that locus. The question for a realization is which bearer each basis tracks and whether the proposed preservation and breakage judgments are warranted.

For scopes, the proposed extension-fixed and structure-fixed bases are compared using aggregation and decomposition (Proposition 5.5). An operation may preserve coverage while changing component structure. Whether that difference is an identity difference depends on the independently justified criterion for the scope in question.

For rules, the proposed content-fixed and structure-fixed bases are compared using refinement (Proposition 5.6). A refinement may preserve specified content while changing structure. Again, the relevant sameness judgment depends on what the rule is and on which criterion of its identity is justified.

Two justified identity relations that disagree on an eligible pair of presentations are distinct on that comparison domain. A difference between declared transformation classifications does not by itself exhibit a pair on which the

generated relations disagree. Such a witness requires a shared eligible comparison domain and appropriate transformation and identity interpretations. These pairs supply candidate distinctions for further testing.

7.2 Profiles and Generated Relations

The profiles catalogued in Section 6 arise from the selected carrier–basis assignments. Proposition 6.2 concerns distinctions among their classification functions.

The number of distinct family-level generated relations depends on the generating rule. The PRS-only and PRS-or-NEU constructions induce different patterns of equal family sets; Appendix A enumerates them and gives the resulting upper bounds. Further equalities may arise when the dynamics do not distinguish the selected family sets. These bounds do not apply to the invariant-based step relations, whose equality and separation are characterized by Theorem 6.4 and Proposition 6.5.

7.3 Carrier-Indexed Combination

An elementary construction shows that relations on distinct carrier kinds can be represented together while preserving each component equivalence relation. Let K index the selected carrier kinds, and let D_k be a domain of presentations for carrier kind k. Suppose E_k is an equivalence relation on D_k for each $k \in K$. Use the tagged disjoint union

$$D = \bigsqcup_{k \in K} D_k$$

and define a relation E on D by

$$(k, x)\ E\ (\ell, y) \quad \Longleftrightarrow \quad (k = \ell\ \wedge\ x\ E_k\ y).$$

Proposition 7.1 (Carrier-indexed combination)**.** *Then E is an equivalence*

relation on D. Restricting E to the k-th summand recovers E_k for each $k \in K$.

Proof. Reflexivity and symmetry follow from the corresponding properties of each E_k. If $(k, x)\ E\ (\ell, y)$ and $(\ell, y)\ E\ (m, z)$, then $k = \ell = m$ and $x\ E_k\ y\ E_k\ z$. Transitivity of E_k therefore gives $(k, x)\ E\ (m, z)$. For presentations in the k-th summand, the definition of E is exactly the definition of E_k. □

This construction preserves the component equivalence relations without identifying presentations belonging to different carrier kinds. Distinct carrier kinds therefore do not by themselves require separate formal structures: a declared carrier tag carries the distinction within one relation. Whether a combined relation is an adequate identity account, preserves all required transformation semantics, and satisfies the Neutral Substrate constraint depends on the semantics of the tags and on the bases proposed for each carrier kind. A publicly declared carrier or basis parameter is not a hidden regime merely because it is a parameter; its adequacy depends on whether its semantics are fixed and its use is justified.

7.4 Separation Under an Extended Transformation Basis

A narrower monotonicity result applies to the declared classification profiles. Let $\mathcal{F} \subseteq \mathcal{F}'$ be transformation bases. Suppose the classifications of two profiles on $\mathcal{F}'$ agree with their existing classifications on every operation in $\mathcal{F}$.

Proposition 7.2 (Persistence of a classification distinction)**.** *If two profiles have different classifications for some $f \in \mathcal{F}$, their classification functions remain distinct under extensions to $\mathcal{F}'$ that preserve the original classifications.*

Proof. The extended classification functions still disagree at f, so they cannot

be equal. □

This result records that a classification difference, once established, is stable under extensions of the transformation basis that retain the original classifications.

The broader limitations are discussed in Section 11.

8 Determinacy of the Selected Profile Assignment

Sections 4–6 specify a finite catalog of carrier–basis pairs and proposed transformation-classification profiles. This section asks whether the catalog assigns a named profile to each selected pair without ambiguity. This is an internal property of the specified assignment.

8.1 Core Carrier–Basis Pairs

Definition 8.1 (Core carrier–basis pair)**.** A *core carrier–basis pair* is one of the selected carrier-kind and candidate identity-basis combinations displayed below. The arrows record the profile names assigned within this construction:

$$\begin{aligned}
\text{obligation-bearer with responsibility-bearing identity} &\mapsto \text{OBL},\\
\text{occurrence with realization-and-provenance identity} &\mapsto \text{OCC},\\
\text{record with descriptive-record identity} &\mapsto \text{REC},\\
\text{plain referent with locus-fixed identity} &\mapsto \text{LOC},\\
\text{plain referent with object-fixed identity} &\mapsto \text{OBJ},\\
\text{scope with extension-fixed identity} &\mapsto \text{SCOPE-E},\\
\text{scope with structure-fixed identity} &\mapsto \text{SCOPE-S},\\
\text{rule with content-fixed identity} &\mapsto \text{RULE-C},\\
\text{rule with structure-fixed identity} &\mapsto \text{RULE-S}.
\end{aligned}$$

The carrier kinds listed first have one candidate basis each; the others have

two, motivated by the conditional comparisons in Section 5.

Definition 8.2 (Core regime assignment)**.** The *core regime assignment* A maps each selected core carrier–basis pair to the named classification profile indicated by the corresponding arrow in Definition 8.1. Its codomain is the selected profile catalog in Section 6.

The assignment is stipulated as part of the finite construction. It does not choose or justify a basis for an actual referent.

8.2 Existence

Lemma 8.3 (Assignment existence)**.** *Every selected core carrier–basis pair has an assigned profile under A.*

Proof. Definition 8.1 lists all selected pairs and gives an image for each. By Definition 8.2, A is defined on every pair in its stated domain. □

8.3 Uniqueness

Lemma 8.4 (Assignment uniqueness)**.** *Each selected core carrier–basis pair has exactly one assigned profile under A.*

Proof. Each pair occurs exactly once on the left-hand side of the displayed assignment in Definition 8.1, with one profile name on the right-hand side. The definition of A assigns no additional value to that pair. Hence its image under A is unique. □

8.4 Determinacy Theorem

Theorem 8.5 (Core regime assignment)**.** *For each carrier–basis pair in the selected catalog, there exists exactly one named classification profile assigned to it by A.*

Proof. Existence is Lemma 8.3; uniqueness is Lemma 8.4. □

This theorem records determinacy of a defined, finite assignment. It does not derive the assignment from the neutrality constraint, establish the scientific correctness of its classification entries, or show that the assigned profile yields a warranted strict-identity relation on real-world referents. Other candidate profiles or identity bases may be relevant outside this selected catalog.

8.5 Limits of Basis Declaration

A carrier-kind label alone does not select one profile when the catalog contains more than one candidate basis for that kind. For example, a scope may be examined under extension-fixed or structure-fixed criteria, a rule under content-fixed or structure-fixed criteria, and a plain referent under locus-fixed or object-fixed criteria. Supplying the corresponding basis label makes the input to A unambiguous *within this catalog.*

That is not the same as establishing which basis is appropriate for a particular referent. A declared basis needs independently justified individuation and persistence conditions, and each proposed transformation classification needs an adequate interpretation. Even the carrier kinds represented by only one profile in this catalog may admit other legitimate identity bases.

Consequently, a possible conformance check could verify whether a system discloses a selected basis and applies its declared profile consistently. It could not, on that evidence alone, certify the basis's scientific adequacy or the truth of its identity judgments. This is the boundary of the determinacy result proved here.

9 Applications as Identity-Criterion Tests

Sections 4–8 define a selected inventory, candidate carrier–basis profiles, and their stipulated transformation classifications. The following settings illustrate questions that those distinctions can help formulate.

For each setting, the relevant questions are which referent is intended, which presentations are eligible for comparison, what changes under a transformation, and what evidence justifies sameness or difference. Where a common referential judgment cannot be warranted, the Neutral Substrate construction does not license adopting it as an unqualified foundational commitment merely because the records share a label.

9.1 Multi-Jurisdiction Regulatory Reporting

A regulatory record may refer to a corporation, a subsidiary, an office, or a party occupying an office. The selected OBL profile offers a candidate basis for following an obligation-bearer across changes in reported duties and activities. Whether the same legal or institutional bearer persists requires an appropriate continuity criterion; sharing a name, role, or identifier is not sufficient. Different jurisdictions may also identify different bearers for a purported obligation. The shared record must distinguish an identified party from a framework's attribution of responsibility to that party.

A reporting statute is a candidate rule referent. Under RULE-C, the comparison asks whether specified normative content is preserved; under RULE-S, it asks whether a specified textual or internal structure is preserved. Recodification might preserve requirements while changing section organization, yielding different *proposed* classifications. That comparison does not decide whether the resulting enactment is the same statute as a legal object. Its legal continuity, the meaning of its requirements, and the intended unit of reference must be justified independently.

Filings, inspections, transactions, and enforcement actions provide candidate

occurrences for OCC. A filing occurrence is distinguishable from the filed document and from subsequent copies or versions of that document, which can be examined under REC. Two jurisdictions may agree which filing occurred while disagreeing whether it constituted compliance. They may also disagree about whether two reported activities are the same occurrence. In the latter case, common occurrence identity cannot simply be assumed from matching timestamps or document identifiers.

A jurisdictional scope can be examined by the cases it covers (SCOPE-E) or by a specified administrative or procedural structure (SCOPE-S). Dividing a reporting domain into districts may preserve its coverage while changing its partition. This supplies a conditional transformation comparison, not a general rule that the jurisdiction itself persists under subdivision. The identity of a legal jurisdiction may depend on further institutional facts not captured by either profile.

A regulated facility, parcel, or instrument supplies a possible plain referent. The LOC comparison follows a site or location, whereas OBJ follows the object occupying or moving through it. A facility relocated from one site to another need not cease to be the same facility; conversely, continued use of one site does not make successive occupants one object. The proposed basis must match the intended bearer.

These comparisons make all selected carrier kinds available for analysis and illustrate the candidate split pairs. A particular system may need only some of the candidate profiles. Agreement about a carrier's designation does not by itself supply the identity criterion needed to compare it across jurisdictions.

9.2 Legislative History Tracking

A statute can be tracked by its recorded requirements, its enacted text, its section structure, or the legal continuity of the instrument across amendments. The selected RULE-C and RULE-S profiles isolate content- and structure-based comparisons. An amendment may retain specified requirements while

reorganizing sections; a textual continuation may also be assigned a changed interpretation. Neither observation alone establishes the strict identity of the statute before and after the amendment. The referent and its identity conditions must first be fixed.

Enactments, votes, amendments, applications, and judicial decisions are candidate occurrences for OCC. An enactment is not the enacted rule, and an account of an enactment is not the enactment itself. Legislative texts, committee reports, docket entries, and interpretive memoranda supply record candidates for REC. Copies with identical content may be distinct documentary tokens, and later versions may differ without settling whether they remain the same informational object. The record basis must therefore be specified rather than presumed.

Legislatures, agencies, courts, officers, and regulated parties can be considered as obligation- or authority-bearing parties under OBL, provided the intended bearer is distinguished from an office, its occupant, or a particular institutional role. The same office may have successive occupants; the same person may move to another office. Such changes do not, by themselves, settle the identity of either the person or the office.

Geographic, procedural, temporal, and institutional boundaries provide candidate scopes. A redistricting operation can preserve a chosen set of covered cases while changing a partition, offering a comparison between SCOPE-E and SCOPE-S. Objects affected by the legal regime, such as land parcels, facilities, or instruments, may be considered under LOC or OBJ when those bases are appropriate.

Legislative history therefore offers tests for whether distinctions among an instrument, its events, its documents, its institutional bearers, and its scopes can be maintained without adopting contested normative conclusions. The formal profile catalog organizes these tests; it does not decide which legal theory of continuity is correct.

9.3 Cross-Institutional Provenance

Cross-institutional provenance highlights the distinction between a record and what the record purports to describe. A dataset entry, measurement report, transfer log, annotation, or derived metric is a candidate for REC. A copy may preserve content while creating a new token; an annotation or derivation may instead create a new version or informational object. Neither common content nor an unbroken provenance chain alone settles which record-identity criterion applies.

A sampled object, instrument, facility, or collection site is a candidate plain referent. The LOC profile asks about continuity of a specified site or position; the OBJ profile asks about continuity of a specified object. Forking record representations does not necessarily fork the represented object. Conversely, a process that separates, replaces, consumes, or transforms physical material may raise identity questions that preservation of a location or inventory label cannot resolve. The intended referent and transformation effects must be established separately.

Producing organizations, custodians, laboratories, agencies, and recipients supply possible OBL carriers where their continuity and responsibility-bearing status are independently specified. Transfers, measurements, analyses, approvals, and custody changes supply possible OCC carriers. A transfer log is a record of a proposed transfer occurrence, not automatically that occurrence or proof of its identity.

Data-use policies, consent terms, retention rules, and access restrictions provide candidate rule referents for RULE-C or RULE-S. Institutional coverage conditions, consent domains, and data-sharing boundaries provide candidate scopes for SCOPE-E or SCOPE-S. Whether changes to policy text, permitted uses, scope coverage, or organizational structure preserve the relevant referent remains dependent on justified identity criteria.

These distinctions help separate shared reference from disagreements about data quality, classification, causation, responsibility, or permissible use. They

also expose circumstances in which institutions lack enough warranted common reference for the Neutral Substrate assumption to be discharged. In that case, disputed co-reference or persistence judgments should not be treated as established merely because the records can be joined.

9.4 Summary

The settings illustrate the analytic use of the selected carrier–basis catalog: OBL, OCC, and REC organize questions about bearers, occurrences, and records; LOC/OBJ, SCOPE-E/SCOPE-S, and RULE-C/RULE-S organize candidate comparisons between different bases for plain referents, scopes, and rules. Their usefulness lies in keeping these questions explicit. The examples do not independently validate the table's transformation classifications. To assess scientific adequacy, a concrete study must identify the intended referents and eligible presentations, justify its proposed identity criteria, and evaluate the resulting judgments against relevant examples and counterexamples.

Reference kinds select questions for analysis; declared bases propose criteria; transformation profiles state candidate consequences. Strict identity and persistence under transformation require further warrant, and the conditions for combining such criteria without loss remain open.

10 Related Work

This paper is situated at the intersection of formal ontology, provenance, and classification under disagreement. It does not propose a new upper ontology. It compares selected carrier–basis profiles under specified transformations and derives how their classifications and generated relations behave.

10.1 Classification under Disagreement

Classification systems can embed institutional, social, and evaluative commitments rather than functioning as neutral descriptions (Bowker and Star 1999; Haslanger 2012). In scientific and policy settings, disagreement may persist because participants adopt different evidentiary, methodological, or evaluative frameworks, not merely because a temporary empirical uncertainty remains (Longino 1990). These observations motivate shared representations that do not presuppose interpretive convergence.

The present paper takes this motivation from *Neutral Substrates* (Case 2026a). That work establishes a conditional sufficiency result for semantic neutrality under Referential Common Ground; it does not supply independently justified identity criteria for every intended referent. Here the narrower question is how proposed identity bases respond to specified record transformations, and what evidence would justify their use for shared reference.

10.2 Identity, Roles, and Ontological Methodology

Formal ontology distinguishes identity criteria from ordinary classification and emphasizes the need to examine how roles and other properties relate to the identity of their bearers. OntoClean, in particular, analyzes rigidity and identity-related constraints on taxonomic organization (Guarino 1999; Guarino and Welty 2002). Accounts of social and institutional entities further examine bearers, roles, social objects, and institutional structures (Masolo et al. 2003; Guizzardi 2005).

This paper uses those distinctions as motivation, not as a derivation of its proposed identity bases. A role can be relevant to representing or tracking a referent without independently supplying a sufficient criterion for that referent's strict identity. Conversely, the use of an explicit role or basis parameter does not by itself make a representation inadequate. The question is which referent is intended, what conditions justify treating presentations

as the same referent, and what a transformation actually preserves.

The paper compares declared transformation classifications under candidate identity bases. A difference between classifications is a formal distinction, but it does not alone prove a difference between generated equivalence relations or scientifically warranted identities.

10.3 Upper Ontologies

Upper ontologies such as BFO, DOLCE, and UFO provide resources for modeling entities, occurrences, dependence, qualities, roles, and institutional or social objects (Smith et al. 2015; Arp et al. 2015; Gangemi et al. 2002; Guizzardi 2005). They address broader ontology-engineering purposes than the finite comparison undertaken here. The profiles are not offered as additions to those ontologies, which may already make the distinctions.

For example, a BFO-based model might represent a particular occurrence using an occurrent category and a material object using a continuant category, subject to the ontology's actual modeling commitments. Rules, records, and institutional bearers likewise require analysis of the intended referents and the resources available in a chosen ontology. Such mappings cannot be assumed solely from names in the selected inventory. They must be justified in the application concerned.

DOLCE- and UFO-based representations similarly offer resources for modeling social objects, roles, qualities, and dependence relations. A realization built with any of these frameworks can be examined for whether it distinguishes, where appropriate, a rule's content from its structure, a scope's coverage from its organization, a location from an occupying object, and a record from the thing recorded. Whether those are identity distinctions, rather than differences of description or representation, remains dependent on the selected referent and independently justified identity criterion.

Sections 5–7 compare such candidate distinctions. The comparison can

help make modeling assumptions explicit and expose cases needing further scientific analysis.

10.4 Provenance, Versioning, and Exchange

The transformation families used here resemble operations encountered in provenance and versioning systems. PROV provides vocabulary for activities, entities, derivation, attribution, generation, use, and revision (Moreau and Missier 2013). These relations may supply evidence about how representations were produced or changed. They do not, without additional identity conditions, settle whether two records are the same record, copies of a record, or records about the same referent.

An exchange formalism such as Common Logic may provide a setting for expressing some of the relevant commitments (ISO/IEC 2018). Conservative extension and heterogeneous ontology exchange are related logical concerns (Hodges 1997; Kutz et al. 2010). However, compatibility of a substrate with admissible frameworks is not, by itself, a proof of conservativity, and this paper constructs neither a Common Logic encoding nor a conservative-extension theorem. Its formal results concern the selected transformation profiles and the limits of conclusions drawn from them.

Related questions about long-lived reference and reuse arise in information infrastructures and digital-twin systems (Edwards 2010; Grieves and Vickers 2017). The candidate profiles offer questions to test in such settings: what is the intended bearer, which transformations preserve it, and what available evidence warrants those judgments? The paper does not establish a generally applicable solution for those systems.

10.5 Causal and Normative Separation

The distinction between observations or records and causal claims is important in causal analysis: causal conclusions depend on modeling assumptions

beyond a descriptive record alone (Pearl 2009). The neutral-substrate constraint makes an analogous distinction at the foundational layer. A substrate may record that a specified source asserted a causal or normative proposition without itself adopting that proposition as fact. Under the assumptions of *Neutral Substrates*, the permitted foundational restriction is sufficient for semantic neutrality (Case 2026a).

The present paper examines a different issue: whether identity judgments about records, rules, occurrences, and their referents remain justified across transformations. Annotation, copying, refinement, and provenance extension may preserve some features and change others. Which changes preserve a particular referent's strict identity is not determined merely by a transformation-family label. Keeping interpretation attributed also does not establish that the referential commitments themselves are adequately grounded.

10.6 FAIR Reuse under Persistent Disagreement

The FAIR principles motivate discoverable, accessible, interoperable, and reusable data across communities and over time (Wilkinson et al. 2016). They do not, by themselves, establish the referent-specific criteria needed to justify every co-reference or persistence judgment under persistent disagreement. Shared vocabularies, identifiers, schemas, and documented provenance can support reuse, but their presence alone does not prove strict identity.

The finite construction studied here complements these concerns by making selected identity-basis choices and proposed transformation effects explicit. Its candidate profiles can serve as comparison cases where different identity questions arise during reuse. Whether a particular case calls for separate profiles, an explicit carrier-indexed representation, or another construction requires independent analysis.

11 Limits and Boundary Conditions

The results concern a selected finite construction: the reference kinds, candidate carrier–basis profiles, and transformation basis of Sections 3–6. Within it, the paper derives selected classifications from explicit models, characterizes invariant-preserving step-generated relations, and establishes the determinacy of the profile assignment.

11.1 Conditional Scope

The analysis is motivated by the neutral-substrate constraint of *Neutral Substrates* (Case 2026a) and by the setting of persistent interpretive disagreement described in Section 3. The neutrality result in that earlier work is conditional on Referential Common Ground and independently justified referential commitments. Neither the inventory selected here nor its transformation classifications establishes those premises for a concrete system.

The present results are relative to the selected carrier kinds, candidate identity bases, eligible transformation instances, and declared classification assignments. Their mathematical validity within the specified construction must be distinguished from the scientific adequacy of those choices. In particular, a record's stable label, a declared basis, or a non-breaking transformation classification does not, by itself, warrant strict identity across presentations.

The certified classification cells establish consequences of explicit invariant bases, effect models, and warrants. They do not establish that those models adequately describe the transformations or identity criteria of an actual system.

Individual derived cells may rely on different presentation spaces, field decompositions, and admissibility conditions. Their separate certification does not establish a coherent common semantic realization of the entire classification matrix.

Empirical counterexamples can challenge a proposed model or universal verdict. Finite observations alone do not establish universal conformance of an unrestricted implementation to that model.

The analysis is not tied to a particular legal theory, policy framework, or causal account. That independence of vocabulary does not prove that its selected identity conditions are valid in every such domain.

11.2 Completeness and Combination

The catalog is a selection, not a complete inventory of identity bases or referent kinds, and its size is a property of the construction. The single-basis entries do not assert uniqueness. Records may require distinctions among content, version, token, signature, and chain of custody. Occurrences may be analyzed as bounded happenings, phases, or event complexes. Obligation-bearing parties may be distinguished from offices, roles, and organizational units. Whether any such distinction determines identity depends on an independently justified criterion for the intended referent. Extending the transformation basis may reveal further candidate distinctions or inadequacies in previously assigned classifications; such outcomes require their own analysis.

The carrier-indexed combination of Section 7 shows that separate equivalence relations can be represented together. Whether a given combination retains every warranted identity judgment is a question about the identity semantics of the application.

11.3 Limits of the Selected Setting

The following cases expose questions that the selected construction does not settle. They should not be read as blanket exclusions of otherwise legitimate representational methods.

Interpretively dependent identity. A framework may supply a causal, normative, or institutional criterion under which a referent is identified. Such a judgment can be recorded as an attributed interpretation. Treating it as an unqualified shared referential commitment requires independent grounds and compatibility with the applicable admissible frameworks. The present profile catalog does not establish those grounds.

Relation-supported identity. Relations may provide evidence or conditions for individuation, co-reference, or persistence. Their role cannot be ruled out by stipulation. When a construction uses relational criteria, it must explain what they establish, for which referents, and under what interpretation conditions. This paper does not derive a general prohibition on relation-determined identity.

Framework-relative identity judgments. Admissible frameworks may disagree about co-reference or persistence. Recording their distinct judgments does not itself establish a common identity judgment. If the substrate adopts a contested judgment as foundational, the Referential Common Ground premise must be re-examined. Disagreement alone, however, does not prove that all shared reference has failed.

Declared parameters and undeclared distinctions. A role, flag, carrier tag, or identity-basis parameter is not intrinsically a hidden regime. An explicit parameter can preserve a meaningful distinction within a combined representation. Conversely, an undocumented discriminator or a parameter whose meaning varies across participating frameworks may fail to support warranted shared identity judgments. Whether a proposed combination is faithful must be assessed rather than decided by its number of labels.

Unmodeled transformations. The selected transformation basis $\mathcal{F}$ is not complete. An application may involve operations whose effects differ

from the proposed family-level assignments or require finer typing. Such cases limit the adequacy of the current catalog for that application.

11.4 Scope Exclusions

This paper does not construct causal or normative interpretive frameworks, determine which interpretation should govern, or prescribe a universal ontology. It does not provide a method for discovering the correct identity criterion for an arbitrary referent, certify the soundness of transformation classifications in deployed systems, or establish a general procedure for deciding strict identity.

The paper also does not prescribe that every identity-relevant distinction must be implemented as a separate regime object. A construction using explicit carrier or basis parameters may be suitable if its semantics are fixed, its judgments are warranted, and the required distinctions are preserved. A context-relative construction may serve a different objective, provided the limits on shared reference and interpretation are stated. The scientific questions are what the construction preserves and why those preservation claims are justified, not merely where a distinction appears in a schema.

11.5 Boundary with Accountable Records

This paper investigates candidate identity bases and their proposed behavior under transformations of representations. It does not provide a conformance checker for deployed record systems, mandate a schema for basis declarations or transformation logs, or establish that an implementation discloses every identity-relevant mechanism and use.

Related operational-identity and accountable-records work can examine whether implemented behavior agrees with a declared criterion and whether relevant evidence is observable. Such an audit cannot, solely from agreement with a declaration, certify that the declaration supplies a scientifically ade-

quate identity criterion. Conversely, the construction here does not prescribe implementation categories for such an audit.

11.6 Boundary of the Claim

Within the specified finite construction, the candidate profiles have pairwise distinct classification functions, the carrier–basis assignment is determinate, and classification distinctions persist under extensions that retain the original classifications. The model-derived comparisons show how different proposed identity bases assign different effects to an operation family, and the step-generated results give exact conditions under which their relations coincide or separate.

These are formal results about declared bases and models. Whether a basis yields a warranted identity relation for particular referents depends on independent justification of that basis, the transformation models, and their warrants.

12 Conclusion

This paper pairs candidate identity bases with families of record transformations and derives, rather than stipulates, how selected families classify relative to a basis. Invariant-defined bases and explicit effect models yield verdicts, certified in Lean 4 for scope decomposition, rule refinement, and forking, that expose their dependence on field granularity, admissible steps, and continuity warrants.

The identity relation generated by invariant-preserving steps has an exact characterization: two bases generate equal relations on a comparison domain precisely when they agree on the preservation of every admissible step there, and a single separating step witnesses a difference. Under stated assumptions, classification-based relations can be compared with the step-level construction.

The carrier-indexed combination preserves relations on distinct carrier kinds within one tagged relation.

Coverage versus internal structure for scopes, content versus structure for rules, and locus versus object for plain referents serve as conditional tests of the proposed distinctions, with decomposition, refinement, and forking as the testing operations.

These results make identity claims about transforming records auditable, supply testable conditions for systems that transform them, and identify the assumptions behind each verdict. They are conditional on the declared bases, effect models, and warrants. Naming a reference kind or classifying a transformation does not settle the identity of its referents, so applying the results to particular referents requires independently justified individuation, co-reference, and persistence criteria.

The paper takes its motivation from the conditional neutrality result of *Neutral Substrates* (Case 2026a), which gives sufficient conditions for semantic neutrality under Referential Common Ground but does not supply identity criteria for particular referents.

Remaining work is mathematical and scientific: justify the proposed identity bases for their intended referents, validate transformation classifications on eligible presentations, relate generated relations to warranted co-reference and persistence judgments, and determine which distinctions require separate representation. The application settings of Section 9 offer contexts for these tests, and companion operational-identity and accountable-records work can examine agreement between implementations and declared criteria.

Stable reference under persistent disagreement remains the objective.

Statements and Declarations

Acknowledgments

The author thanks the anonymous reviewers of an earlier version of this work for comments that helped improve the framing, organization, and presentation of the work.

Author Contributions

The author is the sole contributor to this work and is responsible for all aspects of the research, authorship, and publication.

Use of AI-Assisted Tools

AI-assisted tools were used for editing, formatting, and consistency checking, and in developing and reviewing mathematical definitions, proofs, and the companion Lean 4 formalization. The author reviewed all suggestions and is solely responsible for the content.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest.

A Pairwise Distinctions and Generated Relations

The declared classification functions of the candidate profiles in Table 1 are different for each pair, as established by Proposition 6.2. This is a result about the *proposed matrix*; distinct classification functions need not generate distinct identity relations.

A.1 Family-Level Generated Relations

Definition 3.5 generates $\sim_\tau$ using admissible steps of families classified PRS or NEU.

This construction selects steps by their stipulated family classifications rather than independently testing invariant preservation at the individual-step level.

Within the stipulated matrix, OBL and REC have identical sets of non-breaking families. Consequently, on a common eligible domain with the same admissible transformation dynamics, this paper's construction generates equal relations for those profiles. This is a formal comparison; the distinct intended carrier kinds have not been shown to share a scientifically warranted comparison domain.

The companion Persistence classification construction differs: it generates relations using only families licensed PRS.

Within the stipulated matrix, OBL/RULE-C and SCOPE-S/RULE-S have identical sets of PRS families. They therefore generate identical relations under that construction on shared transformation dynamics.

Direct enumeration of the stipulated matrix gives seven distinct PRS-only family sets and eight distinct PRS-or-NEU family sets.

These counts bound the number of distinct relations generated on common dynamics by the respective classification-based constructions. They do not count scientifically required regimes and do not bound the invariant-based

step relations in general.

All other pairs have different sets of PRS families and can be separated under suitable dynamics. Such differences do not prove separation on arbitrary dynamics, which may lack steps witnessing those differences.

These PRS-only results do not establish corresponding equality or separation results for this paper's PRS-or-NEU generated relation.

A.2 Invariant-Generated Relations

The newer invariant-based construction includes the admissible individual steps that preserve a declared invariant, rather than selecting entire families solely by their matrix labels.

Theorem 6.4 establishes that two bases on a common effect model and comparison domain generate equal relations exactly when they agree on invariant preservation for every admissible step in that domain.

Proposition 6.5 establishes that an admissible step preserved by one basis and changed by another is a sufficient witness of different generated relations, provided both endpoints lie in the comparison domain.

Conversely, bases that classify a family differently can still produce equal relations on a restricted domain containing no separating step.

None of these three generated-relation constructions establishes strict identity without an independently justified interpretation.

A.3 Conditional Examples

The scope, rule, and fork models discussed in Section 6.1 demonstrate how explicit presentation fields, transformation effects, and warrants determine derived verdicts. The fork model is especially instructive: a bare fork does not entail either a change of locus or a change of object. The proposed LOC/OBJ

distinction therefore requires additional continuity warrants. Whether the resulting generated relations coincide or differ also depends on the admissible steps and comparison domain.

The stipulated matrix establishes pairwise differences among the proposed classification profiles. These differences do not necessarily imply distinct generated relations. The invariant-based step results provide explicit conditions for relation equality and separation, while their interpretation as strict identity requires independent scientific justification.

References

R. Arp, B. Smith, and A. D. Spear. *Building Ontologies with Basic Formal Ontology*. MIT Press, 2015.

G. C. Bowker and S. L. Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.

D. M. Case. *Neutral Substrates: A Design Constraint for Shared Records Under Persistent Interpretive Disagreement*. arXiv preprint arXiv:2601.14271, 2026.

D. M. Case. *SE Theory: Identity Regimes* (Lean 4 formalization), release v0.7.0. `https://github.com/structural-explainability/se-theory-identity-regimes`, 2026.

P. N. Edwards. *A Vast Machine: Computer Models, Climate Data, and the Politics of Global Warming*. MIT Press, 2010.

A. Gangemi, N. Guarino, C. Masolo, A. Oltramari, and L. Schneider. Sweetening ontologies with DOLCE. In *Knowledge Engineering and Knowledge Management: Ontologies and the Semantic Web (EKAW 2002)*, volume 2473 of LNCS, pages 166–181. Springer, 2002.

M. Grieves and J. Vickers. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary Perspectives on Complex Systems*, pages 85–113. Springer, 2017.

N. Guarino. The role of identity conditions in ontology design. In C. Freksa and D. M. Mark, editors, *Spatial Information Theory: Cognitive and Computational*

Foundations of Geographic Information Science (COSIT '99), volume 1661 of LNCS, pages 221–234. Springer, 1999.

N. Guarino and C. A. Welty. Evaluating ontological decisions with OntoClean. *Communications of the ACM*, 45(2):61–65, 2002.

G. Guizzardi. *Ontological Foundations for Structural Conceptual Models*. PhD thesis, University of Twente, 2005.

S. Haslanger. *Resisting Reality: Social Construction and Social Critique*. Oxford University Press, 2012.

W. Hodges. *A Shorter Model Theory*. Cambridge University Press, 1997.

ISO/IEC. Information technology — Common Logic (CL): A framework for a family of logic-based languages. ISO/IEC 24707:2018, 2018.

O. Kutz, T. Mossakowski, and D. Lücke. Carnap, Goguen, and the hyperontologies: Logical pluralism and heterogeneous structuring in ontology design. *Logica Universalis*, 4(2):255–333, 2010.

H. E. Longino. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press, 1990.

C. Masolo, S. Borgo, A. Gangemi, N. Guarino, and A. Oltramari. *WonderWeb Deliverable D18: Ontology Library*. IST Project 2001-33052 WonderWeb, 2003.

L. Moreau and P. Missier, editors. *PROV-DM: The PROV Data Model*. W3C Recommendation, 30 April 2013.

J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

B. Smith et al. *Basic Formal Ontology 2.0: Specification and User's Guide*. Technical specification, 26 June 2015.

M. D. Wilkinson et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016.